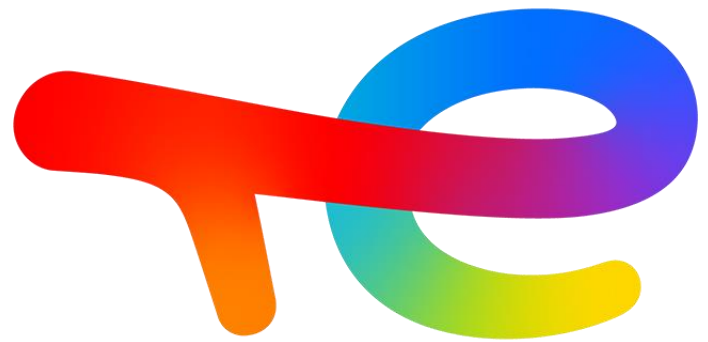




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Performance Contract 2023 - SOUTH-EAST - Objective 12.1
SHAH HABSHAN 6 SOUR OIL DEVELOPMENT

Surface WS

June 23rd, 2023

Agenda

- Introduction, SOW
- Objective History
- Subsurface & Surface analysis
- Conclusions & recommendations

12.1 SHAH HABSHAN 6 SOUR OIL DEVELOPMENT:

Q2 focus on optimal development plan: well design, technologies, surface facilities and tech. costs



Obj.	Description	Q1	Q2	Q3	Q4
12.1	SHAH HABSHAN 6 Sour oil development: - Perform a Prospect integrated study with existing data and with further de-risking activities - Technical cost estimation, proposing technologies & materials of wells and surface facilities.	SoW • Kick-off meeting • Review the data and de-risking activities. • Review high level surface scope with Engineering • Complete understanding on subsurface uncertainty • Define suitable well technologies • Define optimum recovery mechanism	• Screening the surface facilities options • Economic evaluation for various development options	• Conclusion and recommendation	
		DELIVERABLES • KOM with MOM • Finalize the de-risking activities plan • Feedback and structured way forward	Q2 WS • WS with MoM • Profiles and economic analysis sheet • Close out WS	• Final WS and cFDP Report	

Economic evaluation and profiles provided by AON

Part of Q1 WS with MOM

No WS (Final report only)

Knowledge Transfer Planning

- Understanding of best practices for sour oil development across Habshan reservoirs
- Further implement of lesson learned to other undeveloped sour oil (ASAB BQ, BU,...) under UFR

Expected Challenges / issues

- Subsurface uncertainty in volumes, reservoir and fluid.
- Sustainable rates in production and injection wells
- Synergize and expand on the existing surface facilities

Action required to mitigate the expected challenges

- Defining uncertainty envelop by integrating existing data, planned re-risking activities and further proposal (if required)
- Cost effective development scheme, implementation of surface and subsurface technologies
- Detail Facilities Engineering and HSE study

Asset Lead FP

Remi Calvier

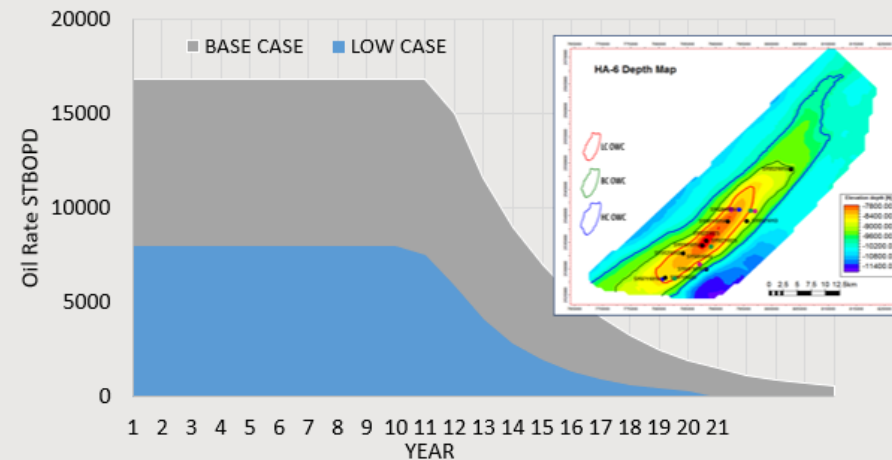
Asset FP

Fatmah Al Yammahi

TC FP

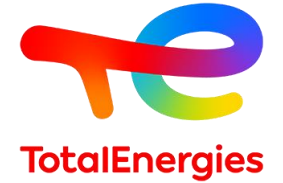
Amer Elbekshi

SHAH HA-6 PRODUCTION PROFILE



Ref: Feb 7th, 2023

SE Objective 12.1 History



Meetings & WS

- ✓ Jan 26: Initial discussion (Abu Dhabi)
- ✓ February 6 : Scope of Work Framing
- ✓ February 13: KOM
- ✓ March 23: Subsurface Peer Review
- ✓ *March: 17th PDISC (with ISH)*
- ✓ April 5: Surface Meeting
- ✓ April 6: Subsurface WS
- ✓ June 6 : UFR roadmap WS (with ISH)
- ✓ June 21 : Drilling WS
- ✓ June 23 : Surface WS.

Input data

2021

- ✓ UFR 2021 ROADMAP WORKSHOP. SULPHUR DEPOSIT RISKS 1 SOUR RESERVOIRS BLENDING OPTIONS STUDY UPDATE, 8-9 March
- ✓ UDR SHAH ROADMAP WS
- ✓ UFR ROADMAP WORKSHOP - SOUR RESERVOIRS BLENDING STUDY UPDATE, September 2021.
- ✓ UFR WORKSHOP - SHAH UNDEVELOPED RESERVOIR, September 2021.

2022

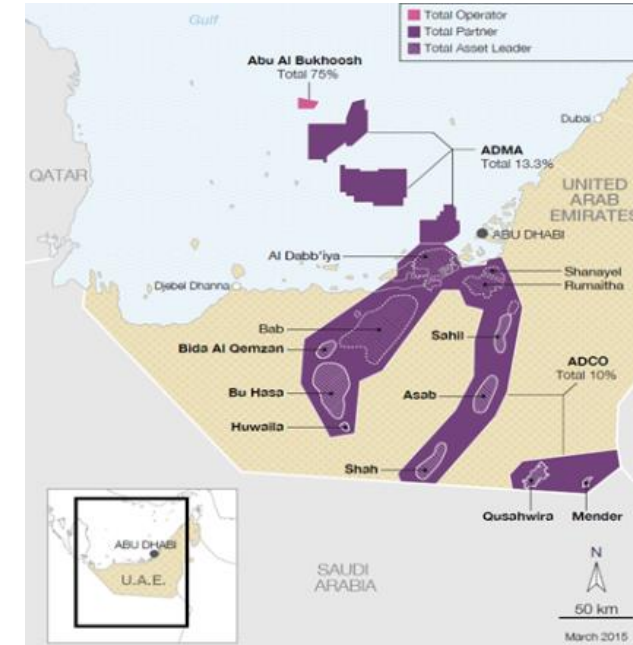
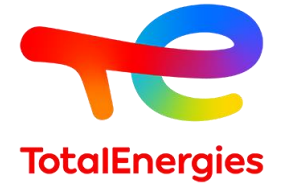
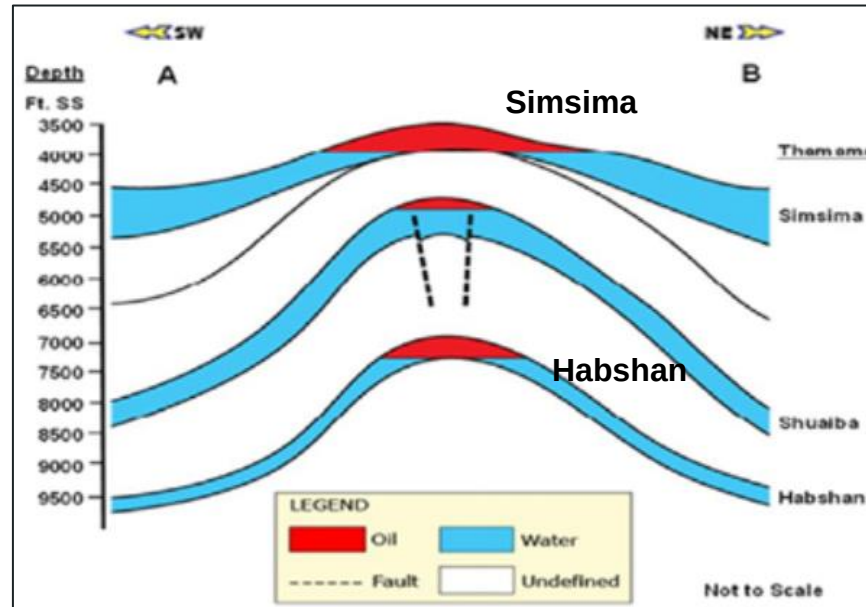
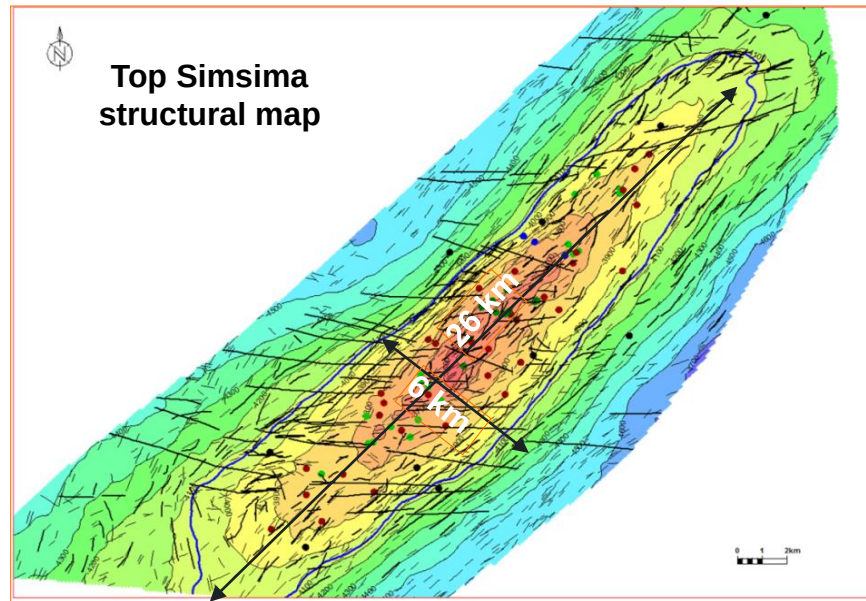
- ✓ UFR 2022 SHAH ROADMAP WORKSHOP (pdf)
- ✓ SOUTH-EAST P5 IPR1 Presentation: SE_P5_IP1_TC_SURFACE_FACILITIES.pdf
- ✓ BP 2023-2027 report

2023

- ✓ March 20: Subsurface and Drilling data
- ✓ April 13: Surface Engineering data
- ✓ May 30 : SHAH Ha-6 analog data
- ✓ June 22 : Deterministic volumes

Shah Field

- Shah field has been producing from the Simsima reservoirs since 1983
- Simsima reservoir
 - Oil : Sweet - muoil 2,3 - 30,5 API
 - Reservoirs R1,R2
 - Perm. : 5 mD – 1 Darcy
 - Thickness : 80 -100 ft
 - Depth : 3600 ft
- Ha-6 reservoir (**UFR**)
 - Oil : 14,1 mole% H₂S - muoil 0,2
 - Reservoir Habshan-6
 - Perm. : 4 mD
 - Thickness : ~25 ft (net 13 ft)
 - Depth : 7500 ft



Challenges of Ha-6 reservoir development : sour oil in tight, thin and deeper reservoir

Appraisal History & Program

- *Habshan-6*
 - 17 well penetrations
 - 1st appraisal well drilled in 2013
 - SY-156V appraisal well in 2022
 - 12 conventional log suites, 3 MDT/RFT
 - 3 cored wells, 1 MICP plugs, 1 SCAL (m&n)
 - 4 Prod.Tests/DST, 1 PVT
 - Main uncertainties: OWC & HC presence in northern area

Reservoir	Reservoir Type	Porosity	Permeability	Net Thickness	STOOIP MMSTB		
		(PU)	(md)	Ft	P90	P50	P10
HA-9	Sour oil/5.89	14 (range: 4 – 26)	1 (range: 0.01 – 21)	14.0	9	24	45
HA-7	Sour oil/10.28	11 (range: 4 – 24)	0.3 (range: 0.01 – 23)	11.0	9	26	46
HA-6	Sour oil /14.14	12 (range: 4 – 23)	1.1 (range: 0.01 – 54)	13.0	379	615	835

MAIN STAKES IN HA-6 RESERVOIR

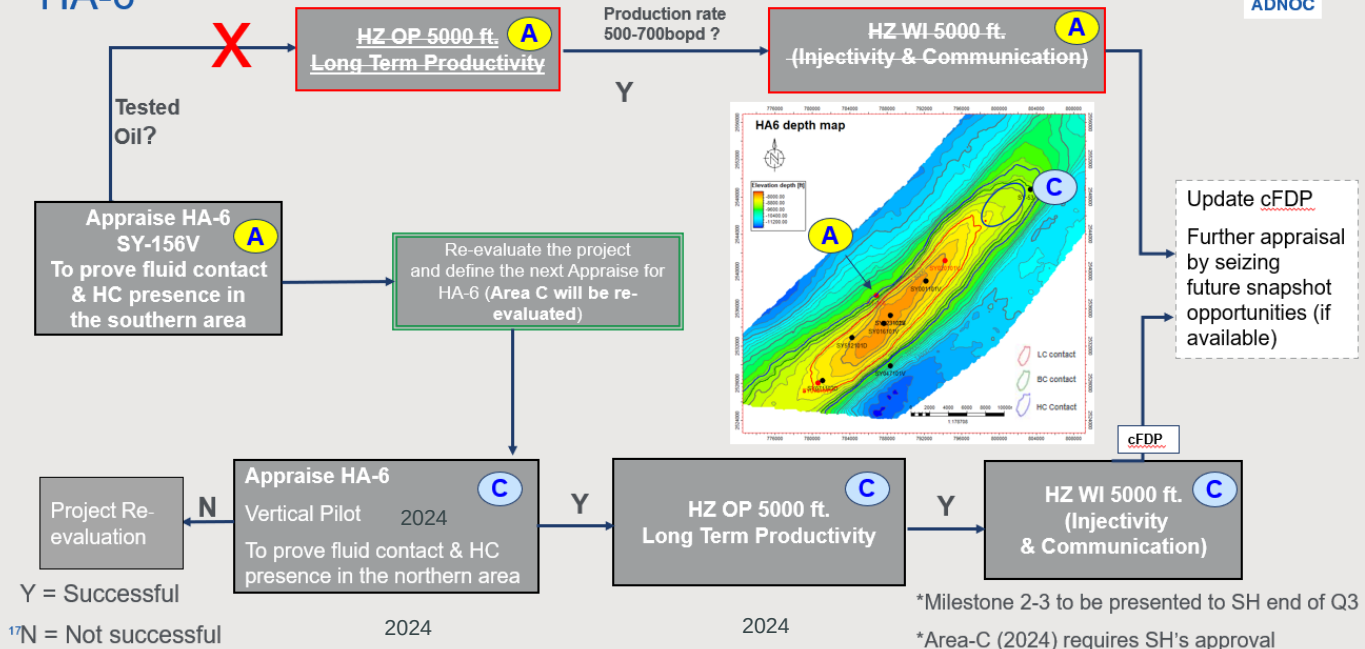
Field Reservoirs	Activity	2021				2022				2023				2024				2025				2026				2027				2028			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4				
Shah HA-6, HA-7 & HA-9	Milestone									1/2/3		4/5/6																					
	Drilling					1 AP				1 AP				1 AP + 1 HP								2 HP + 2 WI								8HP+9WI			
	VAP				G0									G1				G2					G3	T&A									

SY-169 (HA-9 & HA-7)
Approved well

HA-6
To be approved

UFR Roadmap WS - 6 June 2023.
1.10 SHAH Roadmap 2023

APPRAISAL DECISION TREE HA-6



BOD for Development concept

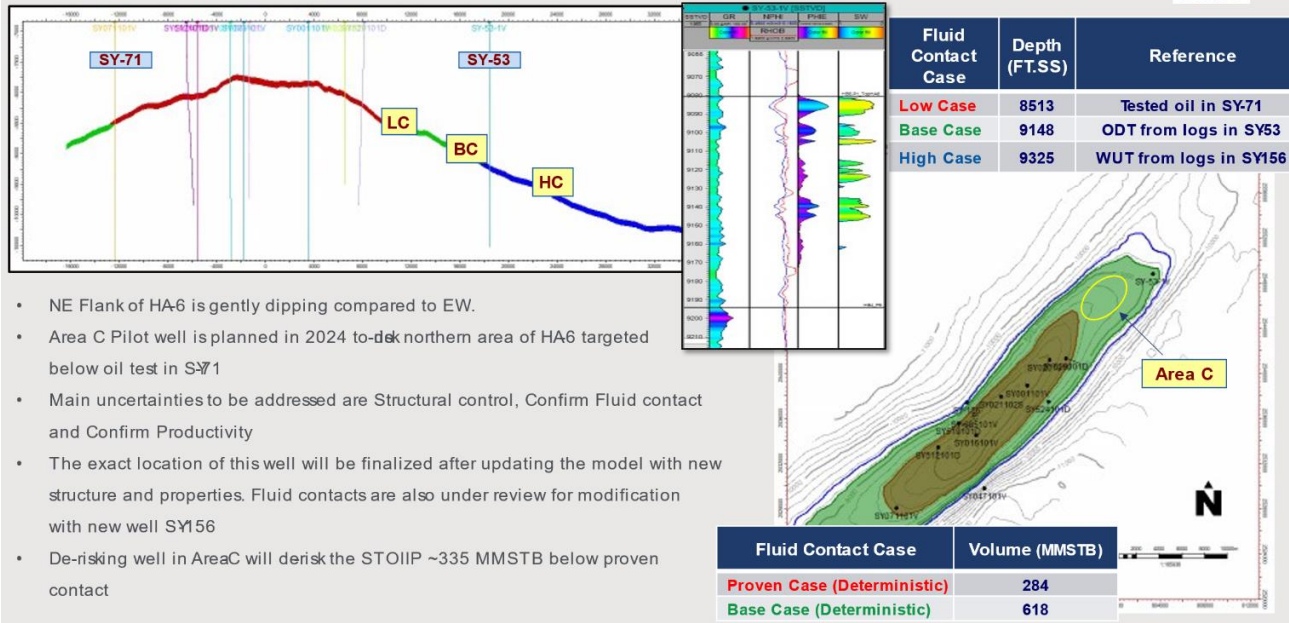
Base case → successful 2024 appraisal

- ✓ STOIIP/ Resources MMSTB: 615 / 185 (615 x 30 % RF)
- ✓ Plateau: 17,5 MBOPD for 20 Years
- ✓ Recovery mechanism: Line Drive Water flood
- ✓ 25 OP & 25 WI
- ✓ First Oil: 2030

Proven case

- ✓ STOIIP/ Resources MMSTB: 284 / 85 ? (284 x 30 % RF)
- ✓ Plateau at ~ 8 MBOPD ?

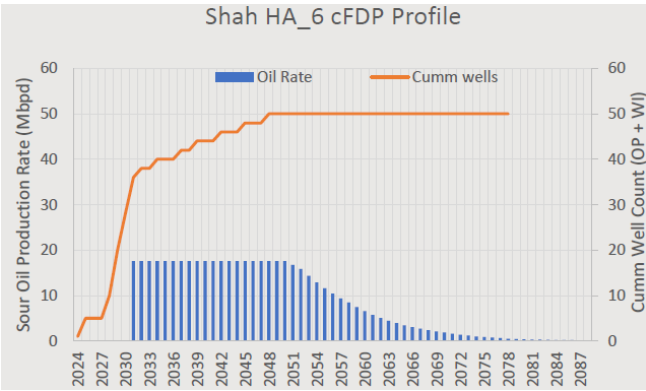
SHAH HA-6 AREA-C DE-RISKING



Clarification 22 June 2023

Table to filled out for subsurface ref (Latest BP)

		Base	High (upside)
STOIIP		615 MMSTB	
RF		30 %	
Plateau rate (number of wells)		17.5 MBPD (25 prod & 25 INJ)	
Plateau length		20 Years	
COP		Line drive with water injection	



SHAH HA-6 Analog data.pdf

Blending study outcome / Adequacy check



Gas processing and Hydraulic study (2022)

- « Quick and high level adequacy with many assumptions »
- UFR load accomodated « while strechting operating conditions limitations »

Risks and uncertainties

- Possibility to produce above 11 MBOPD
- Full Assessment of SOW and costs considering the results of the 2023 and 2024 appraisal campaign
- ASG tarif not yet defined : extra UTC for AON SH.

Adequacy Study Report			
AON Sour Oil Undeveloped Reservoir Fluids processing at Shah Gas Plant Projects & Technical Centre			
A soft copy of this document, maintained on the ADNOC Sour Gas Intranet, is the controlled version. When printed, this document is considered uncontrolled.			
Prepared By:	Eman Al Ali - Process Engineer Jawwad Kaleem - Sr. Process Eng. Sama Alhallaq Sama Alhallaq - Sr. Process Eng.	Document No:	CP04-TS-REP-PEG-0119
Reviewed By:	Muna Almasaghi Muna Yousef Al Maazmi Team leader, Process Eng. - P&TC	Revision No.	0
Approved By:	Ravi Srinivas Manager, Process Engineering- P&TC	Date:	25-Feb-22

BLENDING STUDY OUTCOME

Following rates of Sour Reservoirs can be processed in existing facilities

Field	Sour UFR Routing Option	Reservoir Potential (MBD)	Accommodable Production (MBD)	New Technical Rate	Feasibility Rating
Shah	Shah to ASG	11	11	NA	

Following wells and tie-in based on nFDP for cost estimation Hyp: 10 MMSCFD / H2S gas at 20%

Field	Reservoirs	On Plot Scope	Off Plot Scope
Shah	Habshan	Not applicable in Scenario-A	Well Count (54)
			Oil (ESP) Producers (On Pad wells) = 27
			Water Injector (Off Pad Wells) = 27
			Pads / Clusters = 3 (14 OP wells/PAD)
			WIC = 1
			Pipelines = 136 km
			OHL = 15 km

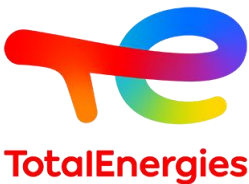
ECONOMICS MODEL UFR SHAH SOUR OIL TO ASG FACILITIES



Max Prod Rate		Volume		CAPEX (\$MM)				OPEX	Output		UTC @ 9%	UTC @ 0%	NPV @ 9%	NPV @ 0%	IRR	PI @ 9%	Economic Cut Off	Payout from
		Oil	Gas	Drilling	Facilities	Abandon	NPC		Number of wells									
Mbd	MM bbl	Bn scf		(\$MM)	(\$MM)	(\$MM)	(\$MM)	(\$MM)	#		(\$/bbl)	(\$/bbl)	(\$MM)	(\$MM)	(%)	Ratio	Year	Months
11	144	122		413	516	142	524	659	0		27.02	10.05	1,031	10,802	19%	1.97	2072	126
			Real Costs	324	458	54		379										

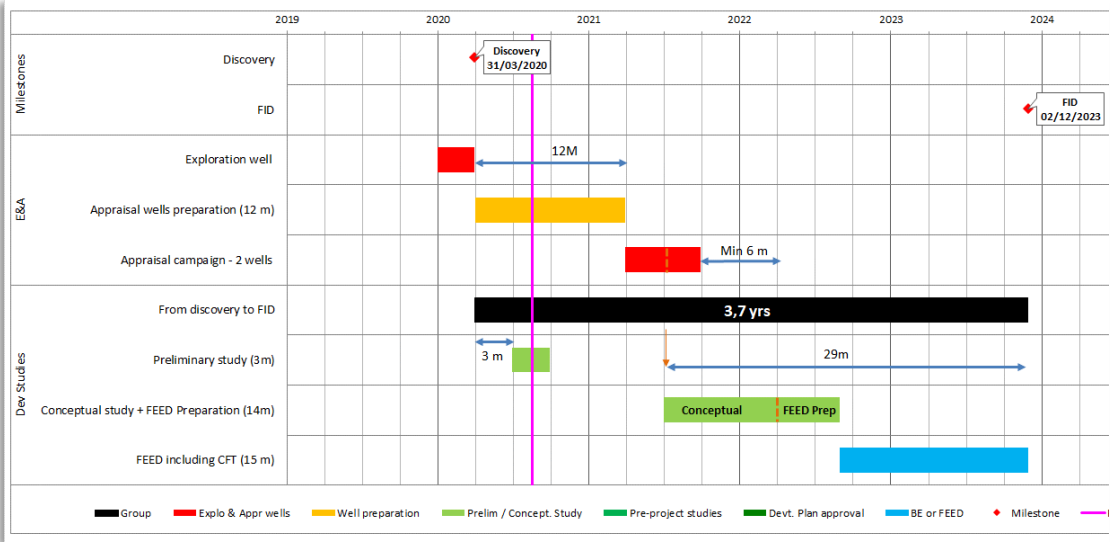
Production plateau : 2032-2054

Development planning



Conceptual study to contribute to optimal development concept with lowest UTC.

- ✓ As a general rule, conceptual study starts when:
 - Subsurface studies are mature enough to provide a consistent reservoir development plan (well count/location, production profiles)
 - Proven and possible resources have a reasonable chance to lead to an economic development.
- ✓ At the end of the conceptual study, the overall development concept is frozen. Any additional reservoir appraisal should not have any significant impact on the selected development scheme.
- ✓ As a reference, one should consider that Conceptual study will not start before the first appraisal well results (with DST / Dynamic data) and will finish at least 6 months after the end of the appraisal campaign allowing a robust concept selection.

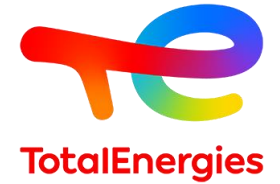


Example of integrated planning

Field Reservoirs	Activity	2021				2022				2023				2024				2025				2026				2027				2028			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Shah HA-6, HA-7 & HA-9	Milestone									1/2/3		4/5/6																					
	Drilling					1 AP				1 AP				1 AP + 1 HP								2 HP + 2 WI								8HP+9WI			
	VAP				G0									G1				G2							G3	T&A							

HA-6 appraisal program to be completed as early as possible to allow G2 in Q1 2025

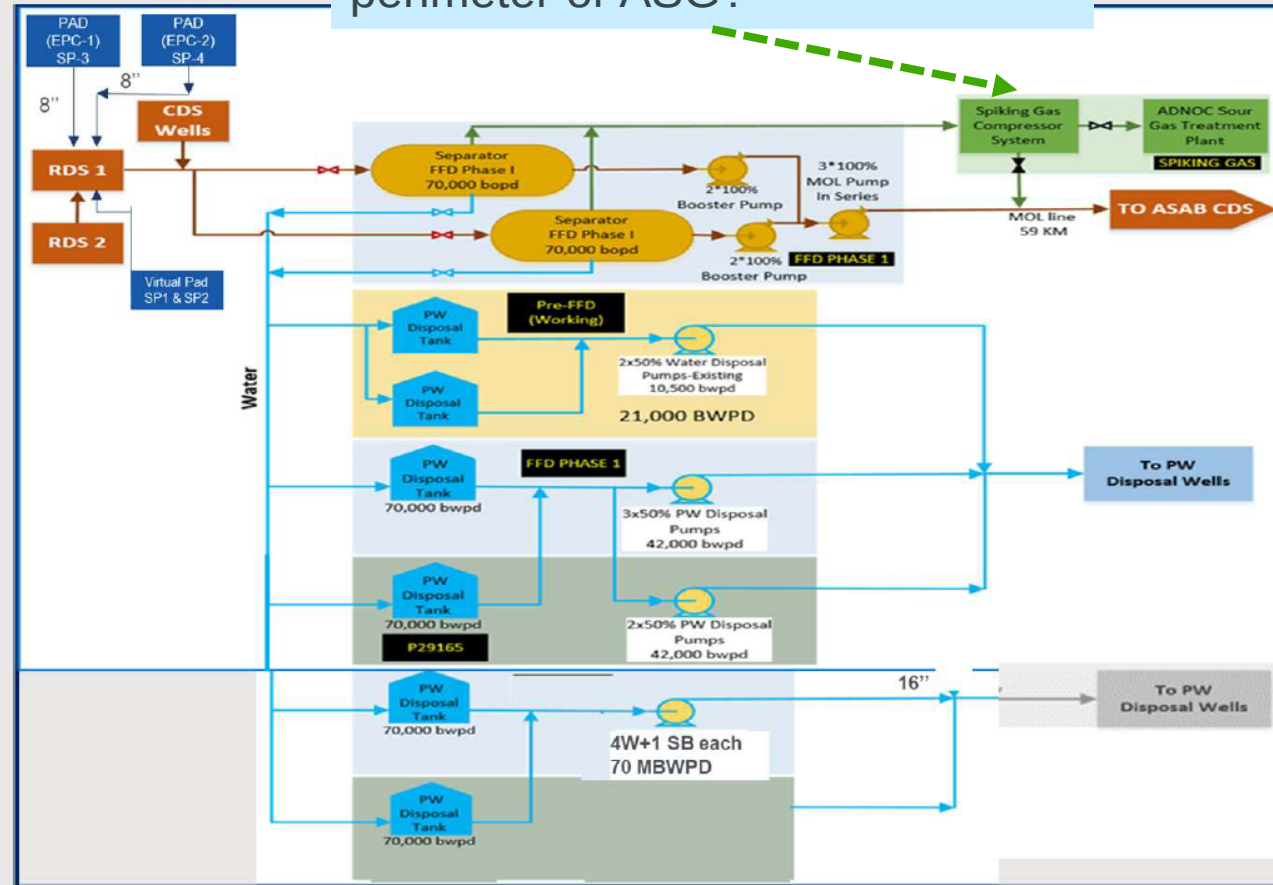
SHAH P5 SURFACE FACILITIES



Shah Offplot Scope :

- 1 new EPC-1 Pad with 900# MSMs/ MPFMs, injection headers, common WHCP(s) and chemical injection packages with utilities systems. Blowdown pipeline are routed from each Pad to nearby connected RDS.
- Additional manifold at RDS-1 with pig receivers for the new transfer lines from new Pads at existing RDS-1.
- Rerouting of 4 existing oil producers is required to connect them to the designated pads.
- New OHLs.
- Tie-in to project wave water injection network.
- No new wellbays in Shah.

Location? Included in SHAH CDS perimeter or ASG?



Bottleneck in spiking gas Compression ?

- Possibility of debottlenecking or adding extra unit in parallel ?

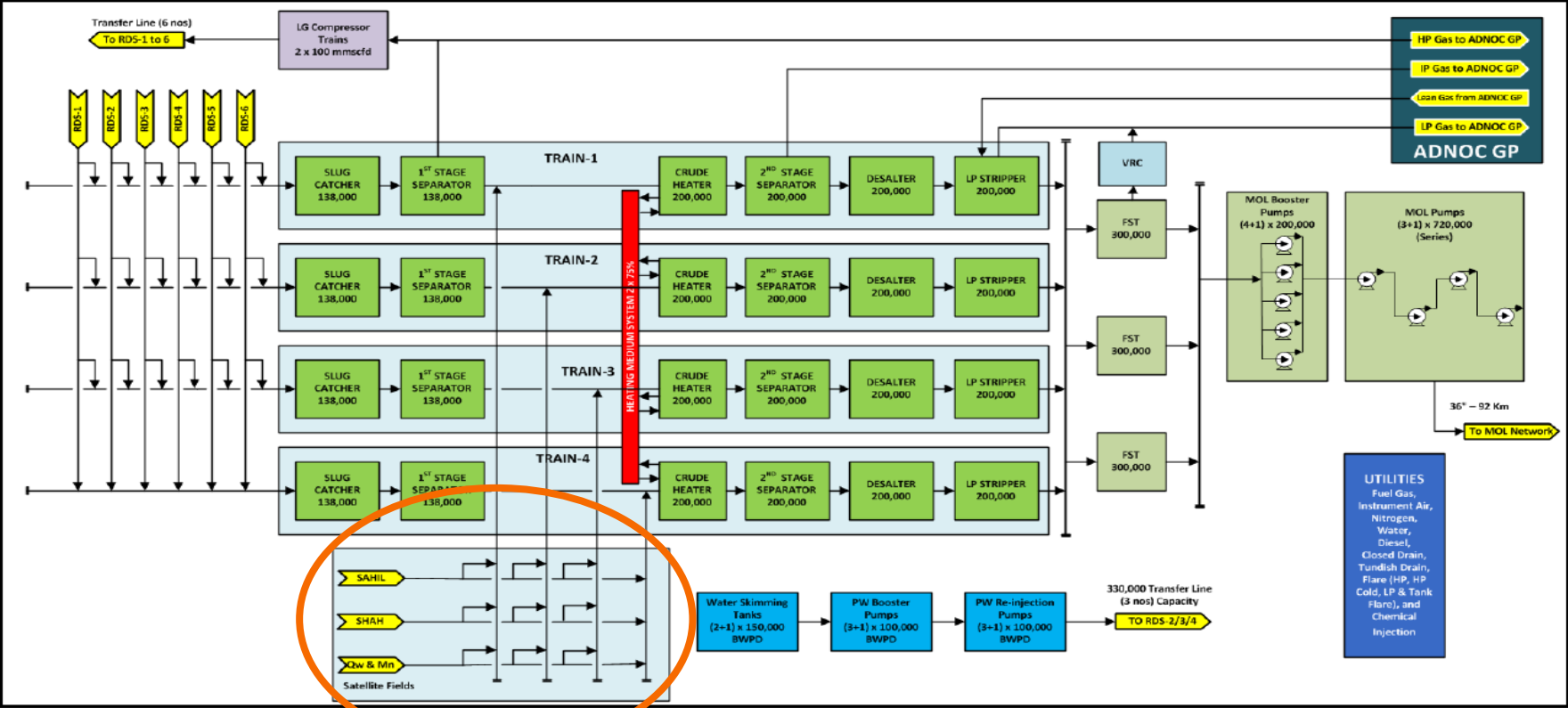
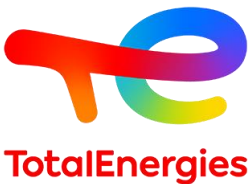
Shah FFD Phase I Separators and booster/MOL pumping systems seem adequate to handle Habshan 6 oil production.

- All the systems working with large redundancy sparing philosophy → to be challenged to increase production ?
- Maximum blending possible ?

Pipelines remaining capacity available ?

Ref: SE P5 IPR1 - Dec 2022 - Surface Facilities

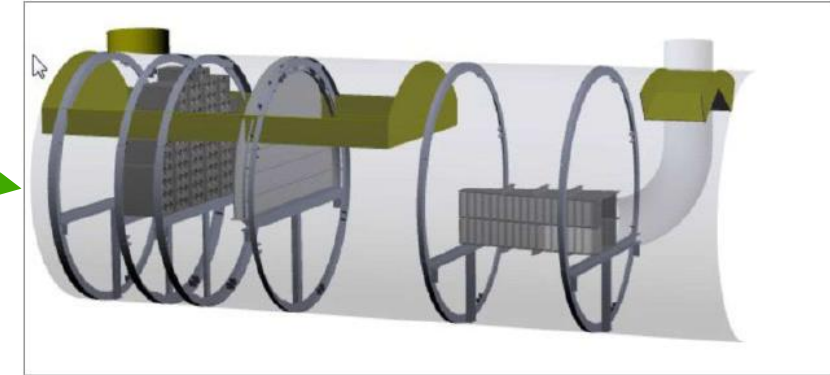
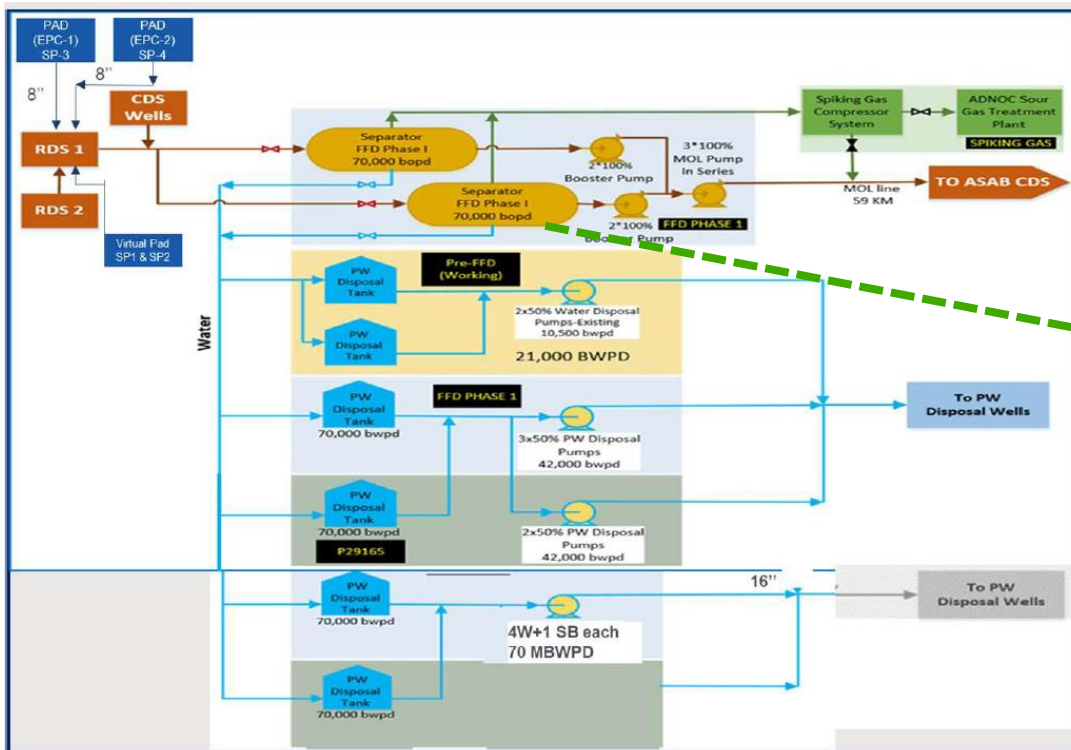
Arrival in ASAB CDS



Shah effluents being handled in a dedicated train in ASAB hub.

What are the maximum blending capabilities there ?

Debottlenecking of oil process train in SHAH



Debottlenecking of Separators in SHAH thanks to the installation of high efficiency internals which could largely manage production increase due to Habshan 6.

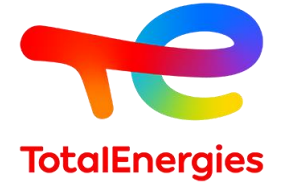
Remaining Water capacity to handle extra water ?

- Question valid for liquid tie-in into Shah or AGS : installation of slug catcher to accommodate possible slugs ?

Conclusions & recommendations

- Habshan-6 is a sour, tight and thin oil undeveloped reservoir currently in appraisal phase.
- 2023 & 2024 remaining appraisal campaign will be key for the definition of the overall development concept.
- Importance to quantify the subsurface stakes and uncertainties, making sure that derisking plan is consistent with the development planning and Select stage.
- Full production processing at Shah gas plant is seen as an option with many uncertainties, high risks and low flexibility.
- Maximizing the utilisation of existing facilities at Shah CDS is recommended to be further studied.
- A carefully planned conceptual study will be key to confirm optimal development concept.
- EOR. Due to the particularity of the field with high concentration of H_2S and CO_2 , a specific EOR screening is recommended based on SCAL data. First action on EOR derisking plan should be the launch of the SCAL campaign with fluids ranging from aquifer water to smart water and gas injection to explore the potential of these fluids towards higher RF. EOR screening is premature without data acquisition.

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